

Chapter 1

What is special about banks?

Learning objectives

- To understand the role of financial intermediaries in the economy
- To understand lenders' and borrowers' different requirements and how banks can help to bridge such differences
- To understand how financial intermediaries reduce transaction, information and search costs
- To analyse the theories of financial intermediation

1.1 Introduction

The first question one may ask when reading this book is 'What is special about banks?' This chapter aims to offer some insights into the nature of the banking business and what makes banks 'special'. A bank is a financial intermediary that offers loans and deposits, and payment services. Nowadays banks also offer a wide range of additional services, but it is these functions that constitute banks' distinguishing features. Because banks play such an important role in channelling funds from savers to borrowers, in this chapter we use the concepts of 'bank' and 'financial intermediary' almost as synonyms as we review the role of banks and their main functions: size transformation, maturity transformation and risk transformation. The difference between banks and other financial intermediaries is introduced in Chapter 2. The second part of this chapter gives an overview of some important concepts in information economics as they apply to banking. The final sections present five theories to explain why banking exists and the benefits of financial intermediation.

1.2 The nature of financial intermediation

To understand how banks work, it is necessary to understand the role of financial intermediaries in an economy. This will help us to answer the question about why we need banks. Financial intermediaries' and financial markets' main role is to provide a mechanism by which funds are transferred and allocated to their most productive opportunities. A bank is

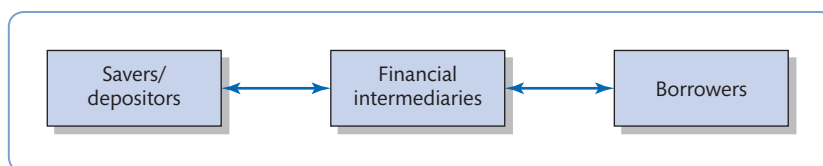


Figure 1.1 The intermediation function

a financial intermediary whose *core* activity is to provide loans to borrowers and to collect deposits from savers. In other words they act as *intermediaries* between borrowers and savers, as illustrated in Figure 1.1. By carrying out the intermediation function, banks collect surplus funds from savers and allocate them to those (both people and companies) with a deficit of funds (borrowers). In doing so, they channel funds from savers to borrowers thereby increasing economic efficiency by promoting a better allocation of resources. Arguably, savers and borrowers do not need banks to intermediate their funds: in **direct finance**, as shown in Figure 1.2, borrowers obtain funds directly from lenders in financial markets. A financial claim is a claim to the payment of a future sum of money and/or a periodic payment of money. More generally, a **financial claim** carries an obligation on the issuer to pay interest periodically and to redeem the claim at a stated value in one of three ways:

- 1 on demand;
- 2 after giving a stated period of notice;
- 3 on a definite date or within a range of dates.

Financial claims are generated whenever an act of borrowing takes place. Borrowing occurs whenever an economic unit's (individuals, households, companies, government bodies, etc.) total expenditure exceeds its total receipts. Therefore borrowers are generally referred to as **deficit units** and lenders are known as **surplus units**. Financial claims can take the form of any **financial asset**, such as money, bank deposit accounts, bonds, shares, loans, life insurance policies, etc. The lender of funds holds the borrower's financial claim and is said to hold a financial asset. The issuer of the claim (borrower) is said to have a **financial liability**. The borrowing–lending process illustrated in Figure 1.2 does not require the existence of financial intermediaries. However, two types of barriers can be identified to the direct financing process:

- 1 The difficulty and expense of matching the complex needs of individual borrowers and lenders.
- 2 The incompatibility of the financial needs of borrowers and lenders.

Lenders are looking for safety and liquidity. Borrowers may find it difficult to promise either.

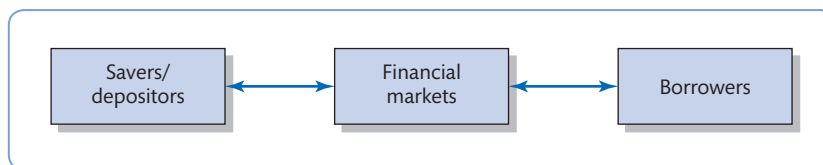


Figure 1.2 Direct finance

Lenders' requirements:

- The *minimisation of risk*. This includes the minimisation of the risk of default (the borrower not meeting repayment obligations) and the risk of assets dropping in value.
- The *minimisation of cost*. Lenders aim to minimise their costs.
- *Liquidity*. Lenders value the ease of converting a financial claim into cash without loss of capital value; therefore they prefer holding assets that are more easily converted into cash. One reason for this is the lack of knowledge of future events, which results in lenders preferring short-term to long-term lending.

Borrowers' requirements:

- Funds *at* a particular specified date.
- Funds *for* a specific period of time; preferably *long-term* (think of the case of a company borrowing to purchase capital equipment which will only achieve positive returns in the longer term or of an individual borrowing to purchase a house).
- Funds at the *lowest possible cost*.

In summary, the majority of lenders want to lend their assets for short periods of time and for the highest possible return. In contrast the majority of borrowers demand liabilities that are cheap and for long periods.

Financial intermediaries can bridge the gap between borrowers and lenders and reconcile their often-incompatible needs and objectives. They do so by offering suppliers of funds safety and liquidity by using funds deposited for loans and investments. Financial intermediaries help minimise the costs associated with direct lending – particularly **transaction costs** and those derived from **information asymmetries** (these concepts will be analysed in more detail in Section 1.4). Transaction costs relate to the costs of searching for a counterparty to a financial transaction;¹ the costs of obtaining information about them; the costs of negotiating the contract; the costs of monitoring the borrowers; and the eventual enforcements costs should the borrower not fulfil its commitments. In addition to transaction costs, lenders are also faced with the problems caused by asymmetric information. These problems arise because one party has better information than the counterparty. In this context, the borrower has better information about the investment (in terms of risk and returns of the project) than the lender. Information asymmetries create problems in all stages of the lending process. Transaction costs and information asymmetries are examples of market failures; that is, they act as obstacles to the efficient functioning of financial markets. One solution is the creation of organised financial markets. However, transaction costs and information asymmetries, though reduced, still remain. Another solution is the emergence of financial intermediaries. Organised financial markets and financial intermediaries co-exist in most economies; the flow of funds from units in surplus to units in deficit, in the context of direct and **indirect finance**, is illustrated in Figure 1.3.

Having discussed the advantages of financial intermediation over direct finance, it is necessary to point out that financial intermediaries create additional costs for borrowers and

¹ Transaction costs can be defined as the costs of running the economic system (Coase, 1937). In particular, it is common to distinguish between co-ordination costs (e.g. costs of search and negotiation) and motivation costs (e.g. costs due to asymmetric information and imperfect commitment). Transaction costs can be measured in time and money spent in carrying out a financial transaction.

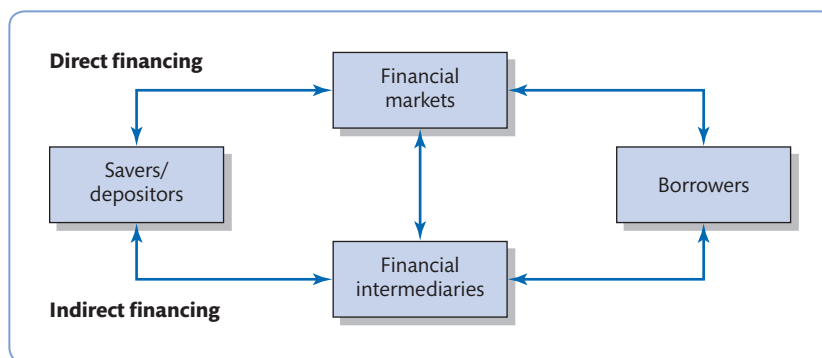


Figure 1.3 Direct and indirect finance

lenders who use their services. Therefore, in order to be able to state that intermediated finance is more advantageous than direct finance, it is necessary that the benefits of such activity outweigh the costs associated with intermediation.

The role of financial intermediation has now become more complex as financial intermediaries perform additional roles, such as brokerage services (i.e. buying and selling stocks and bonds for clients), leasing and factoring. Prior to the 2007–2009 financial turmoil, banks also engaged in a wide process of securitisation (i.e. the pooling and re-packaging of illiquid financial assets into marketable securities) thus creating an extra layer of intermediation, as illustrated in Figure 1.4. When financial intermediaries hold claims issued by other financial intermediaries, then an extra layer of financial intermediation is created. Nowadays, given the increased complexity of credit flows, it is not uncommon to have more than two layers of intermediation.

In the decade leading up to the 2007–2009 financial crisis, financial markets have also witnessed the rapid growth of a different form of financial intermediation, which has become known as **shadow banking**. The term ‘shadow banking’ was first used at the 2007 Jackson Hole Symposium, an annual symposium sponsored by the Federal Reserve Bank of Kansas City. The Financial Stability Board (2011c) defines shadow banking broadly as ‘credit intermediation involving entities and activities outside the regular banking system’. This is, however, a very broad definition and both the scope and the economic relevance of shadow banking is still little understood. This has spurred an academic and policy debate on the role of banks in the financial system, and renewed the need to understand banks’ operations, their

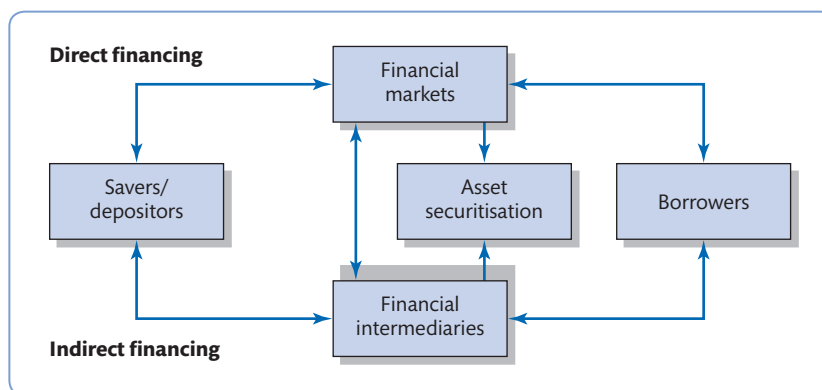


Figure 1.4 Modern financial intermediation

economic role, their risk management systems as well as the activities that are carried out outside the scope of the current regulatory framework. It is widely recognised that the two shadow banking activities that are most important economically and in terms of financial stability at present are securitisation and collateral intermediation (Claessens *et al.*, 2012). These issues will be discussed in more detail in Chapter 18.

1.3 The role of banks

To understand fully the advantages of the intermediation process, it is necessary to analyse what banks do and how they do it. We have seen that the main function of banks is to collect funds (deposits) from units in surplus and lend funds (loans) to units in deficit. Deposits typically have the characteristics of being small-size, low-risk and high-liquidity. Loans are of larger-size, higher-risk and illiquid. Banks bridge the gap between the needs of lenders and borrowers by performing a transformation function:

- size transformation;
- maturity transformation;
- risk transformation.

Size transformation

Generally, savers/depositors are willing to lend smaller amounts of money than the amounts required by borrowers. For example, think about the difference between your savings account and the money you would need to buy a house! Banks collect funds from savers in the form of small-size deposits and repackage them into larger size loans. Banks perform this size transformation function exploiting **economies of scale** associated with the lending/borrowing function, because they have access to a larger number of depositors than any individual borrower (see Section 1.4.2).

Maturity transformation

Banks transform funds lent for a short period of time into medium- and long-term loans. For example, they convert demand deposits (i.e. funds deposited that can be withdrawn on demand) into 25-year residential mortgages. Banks' liabilities (i.e. the funds collected from savers) are mainly repayable on demand or at relatively short notice. Meanwhile, banks' assets (funds lent to borrowers) are normally repayable in the medium to long term. Banks are said to be 'borrowing short and lending long' and in this process they are said to 'mismatch' their assets and liabilities. This mismatch can create problems in terms of **liquidity risk**, which is the risk of not having enough liquid funds to meet one's liabilities.

Risk transformation

Individual borrowers carry a risk of default (known as credit risk), which is the risk that they might not be able to repay the amount of money they borrowed. Savers, meanwhile, wish to minimise risk and prefer their money to be safe. Banks are able to minimise the risk of individual loans by diversifying their investments, pooling risks, screening and monitoring borrowers and holding capital and reserves as a buffer for unexpected losses. The tools and techniques used by banks to perform these transformations and to minimise the risks inherent with such transformations will be illustrated in Chapter 11.

1.4 Information economies

As discussed earlier, banks provide an important source of external funds used to finance business and other activities. One of the main features of banks is that they reduce transaction costs by exploiting scale and scope economies and often they owe their extra profits to superior information. Sections 1.4.1 and 1.4.2 look into information economies as they apply to the banking industry.

1.4.1 Transaction costs

Banks traditionally differ from other financial intermediaries for two main reasons: (1) bank liabilities (i.e. deposits) are accepted as a means of exchange and (2) banks are the only intermediaries that can vary the level of deposits and can create and destroy credit. Modern views on financial intermediation indicate as a crucial function of financial intermediaries the transformation of primary securities issued by firms (deficit units) into secondary securities that are more attractive to surplus units. In this context, financial intermediation can be explained in terms of reduction of transaction costs: secondary securities will be less risky, more convenient and more liquid than primary securities, because banks benefit from economies of scale in transaction technologies and are able to carry out a rational diversification of risks. This allows them to offer lower loan rates relative to direct financing. However, most bank assets are illiquid (non-negotiable) and this can be explained by issues relating to asymmetric information (see Section 1.4.3).

1.4.2 Economies of scale and economies of scope

Financial intermediaries reduce transaction, information and search costs mainly by exploiting economies of scale. By increasing the volume of transactions, the cost per unit of transaction decreases. Moreover, by focusing on growing in size, financial intermediaries are able to draw standardised contracts and monitor customers so that they enforce these contracts. They also train high-quality staff to assist in the process of finding and monitoring suitable deficit units (borrowers). It would be very difficult, time-consuming and costly for an individual to do so. Financial intermediaries can reduce risks by 'pooling', or aggregating, individual risks so that in normal circumstances, surplus units will be depositing money as deficit units make withdrawals. This enables banks, for instance, to collect relatively liquid deposits and invest most of them in long-term assets. Another way to look at this situation is that large groups of depositors are able to obtain liquidity from the banks while investing savings in illiquid but more profitable investments (Diamond and Dybvig, 1983).

Economies of scope refer to a situation where the joint costs of producing two complementary outputs are less than the combined costs of producing the two outputs separately. Let us consider two outputs, Q_1 and Q_2 , and their separate costs, $C(Q_1)$ and $C(Q_2)$. If the joint cost of producing the two outputs is expressed by $C(Q_1, Q_2)$, then economies of scope are said to exist if:

$$C(Q_1, Q_2) < C(Q_1) + C(Q_2) \quad (1.1)$$

This may arise when the production processes of both outputs share some common inputs, including both capital (for example, the actual building the bank occupies) and labour (such

as bank management). Consider, for example, the economies derived from the joint supply of both banking and insurance services. A bank might sell both mortgages and life insurance policies that go with them, therefore creating cross-selling opportunities for the bank. However, the literature indicates that economies of scope are difficult to identify and measure.

1.4.3 Asymmetric information

Information is at the heart of all financial transactions and contracts. Three problems are relevant:

- Not everyone has the same information.
- Everyone has less than perfect information.
- Some parties to a transaction have ‘inside’ information that is not made available to both sides of the transaction.

Such ‘asymmetric’ information can make it difficult for two parties to do business together, and this is why regulations are introduced to help reduce mismatches in information.

Transactions involving asymmetric (or private) information are everywhere. A government selling a bond does not know what buyers are prepared to pay; a bank does not know how likely a borrower is to repay; a firm that sells a life insurance policy does not know the precise health of the purchaser (even though they have a good idea); an investor that buys an equity in Apple does not know the full details about the company’s operations and future prospects. These types of informational asymmetries can distort both firms’ and users’ incentives that result in significant inefficiencies.

Information is at the centre of all financial transactions and contracts. Decisions are made beforehand (*ex ante*) on the basis of less than complete information and sometimes with counterparties who have superior information with the potential for exploitation. In any financial system, information is not symmetrically distributed across all agents, which implies that different agents have different information sets. Put another way, full and complete information is not uniformly available to all interested parties. In addition, not all parties have the same ability to utilise the information that is available to them. In particular, parties have more information about themselves (including their intentions and abilities) than do others. The problem arises because information is not a free good and the acquisition of information is not a costless activity. If either were the case, there would never be a problem of asymmetric information.

Asymmetric information, and the problems this gives rise to, are central to financial arrangements and the way financial institutions behave to limit and manage risk. Information asymmetries, or the imperfect distribution of information among parties, can generate **adverse selection** and **moral hazard** problems as explained in Section 1.4.3.1. Another type of information asymmetry relates to the **agency costs** between the principal (e.g. bank) and agent (e.g. borrower). These issues are analysed in Section 1.4.3.2.

1.4.3.1 Adverse selection and moral hazard

One problem that often arises from asymmetric information is adverse selection. The better-informed economic agent has a natural incentive to exploit their informational advantage. Those who are uninformed should anticipate their informational handicap and behave accordingly. It is the interaction between the inclination of the informed to strategically manipulate and the anticipation of such manipulation by the uninformed that results in

distortion away from the ‘first best’ (the economic outcome in a setting where all are equally well informed). Adverse selection is a problem at the search/verification stage of the transaction (*ex ante*); it is sometimes referred to as the ‘lemon’ problem (Akerlof, 1970). In his famous study entitled ‘Market for lemons’, George Akerlof explains the consequences of asymmetric information in a situation where the buyer, and not the seller, does not know the quality of the commodity being exchanged. In this context, the vendor is aware that they are the only one knowing the true characteristics of the commodities and can exaggerate the quality. Conversely, the buyer can form an opinion on the quality of the commodities only after buying the commodity (i.e. *ex post*). Akerlof demonstrates that if there are a relatively high number of bad commodities in the market, the market will function poorly, if at all. A common example of this phenomenon is in the second-hand car market – here the sellers know whether or not their car is a lemon (a bad car) but the buyers cannot make that judgement without running the car. Given that buyers cannot tell the quality of any car, all cars of the same type will sell at the same price, regardless of whether they are lemons or not. The risk of purchasing a lemon will lower the price buyers are prepared to pay for a car, and because second-hand prices are low, people with non-lemon cars will have little incentive to put them on the market.

One possible solution to the adverse selection problem is to offer a warranty, as it would be viewed as a signal of quality. Hence ‘**signalling**’ refers to actions of the ‘informed party’ in an adverse selection problem. The action undertaken by the less informed party to determine the information possessed by the informed party is called ‘**screening**’ (for example, the action taken by an insurance company to gather information about the health history of potential customers).

Economic transactions often involve people with different information. In the context of financial markets, for example, those who buy insurance or take out bank loans are likely to have a better idea of the risks they face than the insurance company or bank. As such, it is often those who face the bigger risks who are more likely to want to buy insurance and those with the riskiest business proposals who are more likely to seek bank loans and therefore are more likely to be selected. Adverse selection in financial markets results in firms attracting the wrong type of clients; this in turn pushes up insurance premiums and loan rates to the detriment of lower-risk customers. Financial firms such as banks and insurers, therefore, seek to screen out/monitor such customers by assessing their risk profile and adjusting insurance premiums and loan rates to reflect the risks of individual clients.

In banking, adverse selection can occur typically as a result of loan pricing. As shown in Figure 1.5, the relationship between the return the bank can expect from a certain loan and the loan price is increasing and positive up to a certain point (for example, an interest rate of 6 per cent). Any prices above that level (shaded area in the figure) will decrease the expected return for the bank because of adverse selection: only the most risky borrowers (i.e. those with a low probability of repayment, such as speculators) will be ready to accept a loan at a very high price.

Another issue relating to information asymmetries is moral hazard (or hidden action). Superior information may enable one party to work against the interests of another. In general, moral hazard arises when a contract or financial arrangement creates incentives for parties to behave against the interest of others. For example, moral hazard is the risk that the borrower might engage in activities that are undesirable from the lender’s point of view because they make it less likely that the loan will be repaid and thus harm the interest of the lender. A classic example is the use of funds originally borrowed for a ‘safe’ investment project

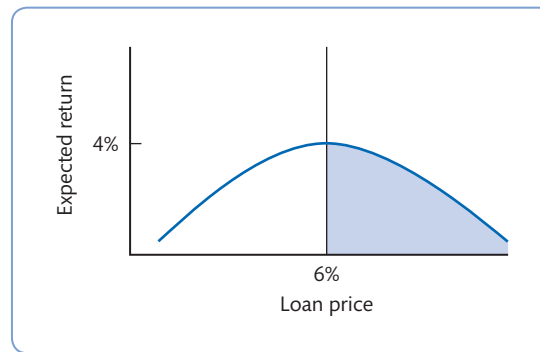


Figure 1.5 Adverse selection in loan pricing

(a car purchase, a home improvement), which are then gambled in a high-risk project (for example, invested in a ‘get rich quick’ scheme). Thus for a bank, moral hazard occurs after the loan has been granted (*ex post*) and is associated with the monitoring and enforcement stages. Those that obtain some form of insurance may take greater risks than they would do without it because they know they are protected, so the insurer may get larger claims than expected. Examples of moral hazard in banking relate to: deposit insurance and the lender of last resort function of the central bank (see Chapter 6).

Monitoring is required whenever there are the problems of moral hazard and adverse selection. A standard example often given is the case of bank loans where lenders screen out excessively high risks, and regularly monitor the performance of the borrowers by obtaining various types of financial information – for example, companies are often required to submit periodic reports detailing the performance of their business. In addition, for loans to large companies, there are credit rating agencies (like Standard & Poor’s and Moody’s) that provide information on firm performance and credit ratings, which is an estimate of the amount of credit that can be extended to a company or person without undue risk (See Table 4.4 Credit risk ratings: Moody’s and Standard & Poor’s). Banks also send inspectors to firms to monitor their progress. However, it is particularly difficult for consumers/investors to monitor financial firms to see how they are performing and what they are doing with their deposits or investments. This is one reason why we have regulators to monitor financial firm behaviour.

1.4.3.2 Principal–agent problems

Financial transactions often create **principal–agent problems** of one sort or another. This is also related to the problem of incentive structures in that the central issue is how a principal is able to rely on the agent acting in the interests of the principal employing them rather than their own selfish interest and against those of the principal. The problem arises because the agent often has superior information and expertise (which may be the reason the principal employs them). The agent can choose their behaviour after the contract has been established, and because of this the agent is often able to conceal the outcome of a contract. Agency problems also arise because the agent cannot be efficiently or costlessly monitored. Unless these problems can be solved, the agency costs involved can act as a serious deterrent to financial contracting with resultant losses. The challenge is to create financial contracts or arrangements that align the interests of the principal and the agent.

A typical example of principal–agent problem refers to a situation of separation of ownership and control in a firm. Managers in control (*the agents*) may act in their own interest rather than in the interest of shareholders (*the principals*) because the managers have less incentive to maximise profits than shareholders.² A firm acting in the interest of the shareholders has an incentive to undertake investments that benefit the shareholders at the expense of creditors. However, as observed by Jensen and Meckling (1976) the assumption that managers act in the best interest of the shareholders is questionable. As an agent of the shareholders, the manager can do many things that may not be in the best interest of the shareholders. For example, managers may select low-risk investment projects with a view toward protecting their positions and reputations. To summarise, the principal (the shareholder) is unable to completely control the agent's behaviour. If it were possible to costlessly observe the agent's action, there would be no moral hazard. It is obvious that the principal anticipates the agent's behaviour. Therefore, the principal attempts to design a contract that aligns the agent's incentives with their own.

The example above shows that principal–agency issues are inextricably linked to information asymmetry and moral hazard. The behaviour of contracting parties (counterparties) needs to be monitored after a contract has been agreed to ensure that information asymmetries are not exploited by one party against the interest of the other, and also because frequently a fiduciary relationship (a relationship of trust and confidence) is created by a financial contract. In both cases, parties need to be monitored to ensure that their behaviour is consistent with both their interests. A special characteristic of many financial contracts is that the value (for example, the future returns on an investment; the amount of loan repayments to a bank – some of which may suffer from default; returns on long-term savings products) cannot be observed or verified at the point of purchase, and that the post-contract behaviour of a counterparty determines the ultimate value of the contract. This also creates a need for monitoring. In addition, monitoring is needed because many financial contracts are long-term in nature and information acquired before a contract is agreed may become irrelevant during the course of the contract as circumstances and conditions change. Above all, the value of a contract or financial product cannot be ascertained with certainty at the point the contract is made or the product is purchased. This often distinguishes financial contracts from other economic contracts such as purchases of goods. While the need for monitoring is accepted, it is an expensive activity and parties involved need to balance the costs and benefits of such monitoring.

As the cost of monitoring principal–agent relationships can be expensive and difficult for the market to resolve, public regulatory agencies help perform this task – for instance, they monitor financial service firms to minimise conflicts between principals (financial firms) and agents (customers).

1.4.3.3 The free-rider problem

One general solution to information problems is for those involved in financial transactions to invest in information. However, this is not a costless activity and free-rider problems may emerge as, in some cases, no one party can appropriate the full value of the costly information acquired.

²Note that shareholders in the US are referred to as stockholders. Also it is useful to remember that equities or ordinary shares in the UK are referred to as common stock in the US.

Free-rider problems occur when people who do not pay for information take advantage of the information that other people have paid for. For example, you purchase information that tells you which firms are good and which are bad. You believe the purchase is worthwhile because you can buy securities of good firms that are undervalued so you will gain extra profits. But free-rider investors see that you are buying certain securities and will want to buy the same.

Governments could produce information to help investors distinguish good from bad firms and provide it to the public free of charge. The main drawback is that this action may be politically difficult. Moreover it never completely eliminates the problem. By encouraging bank lending, a bank can profit from the information it produces by making private loans (i.e. avoiding free-rider problems).

1.4.3.4 Relationship and transaction banking

In credit markets one way to overcome agency and adverse selection problems is for the parties to enter a *relational contract*. Relational contracts are informal agreements between the bank and the borrowers sustained by the value of future relationships. Modern financial intermediation theory has emphasised the role of banks as relationship lenders: this is when banks invest in developing close and long-term relationships with their customers. Such relations improve the information flow between the bank and the borrower and thus are beneficial to both parties. If the customer has a 'history' (e.g. they have borrowed previously from the bank over a long period of time), then the bank's screening and monitoring costs will be much lower compared with the cost associated with new customers. Meanwhile, borrowers will find it easier to get future loans at (relatively) low rates of interest.

The benefits of **relationship banking** over **transactional banking** arise from a reduction of agency problems by long-term loan contracts and the use of information reusability over time. Relationship banking improves upon information flow thus mitigating information asymmetries and allows for flexibility.

The literature (see Boot, 2000; Boot and Thakor, 2014) has indicated that relationship banking can be sustained under high competitive pressures. In other words, the informational savings derived from relationship lending can be considered as a primary source of competitive advantage for existing banks over new market participants; this is because by drawing relational contracts, banks can 'isolate' themselves from competition from other banks and/or non-bank financial intermediaries.

Our discussion above seems to suggest that the creation of strong links between banks and companies might be beneficial to both businesses. Indeed in bank-based systems such as Japan and Germany, banks hold equity stakes in companies they lend to and banks in turn have members on the Board of Directors of these companies. In Anglo-Saxon countries (such as the United States and the United Kingdom) such arrangements are typically not allowed. Intense disintermediation process that has characterised the financial and banking markets, coupled with the increasingly common *transaction banking*, has started to challenge the importance of banks as relationship lenders. Transaction banking involves a pure funding transaction where the bank essentially acts as a 'broker'; an example is that of a mortgage loan made by a bank and then sold on to an investor in the form of a security. This process is known as securitisation and it is explained in more detail in Chapter 18. It is obvious that in transaction banking there is no relationship between the parties and no flexibility in the contract terms.

1.5 Why do banks exist? Theories of financial intermediation

There are five theories that explain why financial intermediation (banking) exists. These theories relate to: delegated monitoring; information production; liquidity transformation; consumption smoothing; and the role of banks as a commitment mechanism.

1.5.1 Financial intermediation and delegated monitoring

One of the main theories put forward as an explanation for the existence of banking relates to the role of banks as ‘monitors’ of borrowers. Since monitoring credit risk (likelihood that borrowers default) is costly, it is efficient for surplus units (depositors) to delegate the task of monitoring to specialised agents such as banks. Banks have expertise and economies of scale in processing information on the risks of borrowers and, as depositors would find it costly to undertake this activity, they delegate responsibility to the banks. One of the most relevant studies explaining why banks exist on the basis of contract theory is by Diamond (1984), according to whom **delegated monitoring** on behalf of small lenders provides the *raison d’être* of banking:

An intermediary (such as a bank) is delegated the task of costly monitoring of loan contracts written with firms who borrow from it. It has a gross cost advantage in collecting this information because the alternative is either duplication of effort if each lender monitors directly or a free-rider problem in which case no lender monitors. Financial intermediation theories are generally based on some cost advantage for the intermediary. Schumpeter assigned such a delegated monitoring role to banks.

(Diamond, 1984, p. 393)

Diamond’s study investigates the determinants of delegation costs and develops a theoretical model in which a financial intermediary (typically a bank or an insurance company) has net cost savings relative to direct lending and borrowing. Diamond’s approach is essentially developed around two interconnected factors:

- 1 Diversification among different investment projects; this is crucial in explaining why there is a benefit from delegating monitoring to an intermediary that is not monitored by its depositors.
- 2 The size of the delegated intermediary that can finance a large number of borrowers.

Since usually diversification will increase with the number of bank loans, larger delegated intermediaries will generate higher economies of scale in monitoring and this will allow for greater portfolio diversification than any individual lender could achieve.

One issue that arises, however, relates to who is ‘monitoring the monitor’. Surplus units (depositors) can reduce monitoring expense if the costs of monitoring the intermediary are lower than the costs of surplus units lending direct to borrowers and therefore directly incurring the monitoring costs. As a financial intermediary increases in size, it can commit to offer deposit facilities to surplus units only if the intermediary is undertaking the appropriate monitoring activity.

1.5.2 Information production

If information about possible investment opportunities is not free, then economic agents may find it worthwhile to produce such information. For instance, surplus units could incur substantial search costs if they were to seek out borrowers directly. If there were no banks, then there would be duplication of information production costs as surplus units would individually incur considerable expense in seeking out the relevant information before they committed funds to a borrower. An alternative is to have a smaller number of specialist agents (banks) that choose to produce the same information.

Banks have economies of scale and other expertise in processing information relating to deficit units – this information may be obtained upon first contact with borrowers but in reality is more likely to be learned over time through repeated dealings with the borrower. As banks build up this information (e.g. the knowledge of credit risk associated with different types of borrowers – ‘customer relationships’) they become experts in processing this information. As such they have an information advantage and depositors are willing to place funds with a bank knowing that these will be directed to the appropriate borrowers without the former having to incur information costs.

1.5.3 Liquidity transformation

Banks provide financial or secondary claims to surplus units (depositors) that often have superior liquidity features compared to direct claims (like equity or bonds). Banks’ deposits can be viewed as contracts that offer high liquidity and low risk that are held on the liabilities side of a bank’s balance sheets. These are financed by relatively illiquid and higher-risk assets (e.g. loans) on the assets side of the bank’s balance sheet. It should be clear that banks can hold liabilities and assets of different liquidity features on both sides of their balance sheet through diversification of their portfolios. In contrast, surplus units (depositors) hold relatively undiversified portfolios (e.g. deposits typically have the same liquidity and risk features). The better banks are at diversifying their balance sheets, the less likely it is that they will default on meeting deposit obligations.

1.5.4 Consumption smoothing

The three aforementioned theories are usually cited as the main reasons why financial intermediaries (typically banks) exist. However, studies have suggested that banks perform a major function as consumption smoothers. Namely, banks are institutions that enable economic agents to smooth consumption by offering insurance against shocks to a consumer’s consumption path. The argument goes that economic agents have uncertain preferences about their expenditure and this creates a demand for liquid assets. Financial intermediaries in general, and banks in particular, provide these assets via lending and this helps smooth consumption patterns for individuals.

1.5.5 Commitment mechanisms

Another theory that has developed aims to provide a reason as to why illiquid bank assets (loans) are financed by demand deposits that allow consumers to arrive and demand liquidation of those illiquid assets. It is argued that bank deposits (demand deposits) have evolved

as a necessary device to discipline bankers. To control the risk-taking propensity of banks, demand deposits have evolved because changes in the supply and demand of these instruments will be reflected in financing costs and this disciplines or commits banks to behave prudently (ensuring banks hold sufficient liquidity and capital resources).

1.6 The benefits of financial intermediation

Financial intermediation, as noted previously, is the process of channelling funds between those who wish to lend or invest and those who wish to borrow or require investment funds. Financial intermediaries act as principals, creating financial assets and liabilities.

A wide range of financial institutions are engaged in financial intermediation, including banks, insurance and pension firms, securities houses and others. Many of the services offered by financial institutions also include both intermediation and non-intermediation activities (e.g. payment services, fund management services and so on).

An important distinguishing characteristic of financial intermediation is that financial assets and liabilities are created. In the case of a bank deposit, the nature of the claims and liabilities created is usually straightforward. The depositor has a claim for a given amount of money, perhaps to be repaid on demand, while the bank has a matching liability to repay a given amount of money. If the bank on-lends deposits it has a claim against the borrower for a given amount of money, to be repaid (with interest) at a given point in time in the future. The borrower, naturally, has a liability to repay that sum of money with interest on the specified date.

The significance of financial intermediation within the financial system is best appreciated in terms of the benefits that it generates. These benefits accrue to *ultimate lenders (surplus units)*, to *ultimate borrowers (deficit units)* and to *society as a whole*.

1.6.1 The benefits to ultimate lenders (surplus units)

These may be summarised as follows:

- *Greater liquidity* is generally achieved by lending to a financial intermediary rather than directly to an ultimate borrower.
- *Less risk is involved*, due to the pooling of risk inherent with financial intermediation, the improved risk assessment that intermediaries are able to undertake and the portfolio diversification that can frequently be achieved. This reduction in risk may be reflected in guaranteed interest rates on deposits with a financial intermediary.
- *Marketable securities* may be issued as the counterpart to deposits with a financial intermediary. For example, a certificate of deposit (CD) is a type of time deposit where the bank issues a certificate that a deposit has been made (this is particularly common in the US). The certificate of deposit can then be sold in the market whenever an individual/firm needs cash. Hence depositors, instead of waiting until maturity of the securities, may sell them in the market to regain the cash. This clearly enhances the liquidity of the depositors' funds (in the broadest sense).
- *Transaction costs* associated with the lending process are likely to be reduced significantly, especially where straightforward deposit facilities are utilised.

- The *lending decision is simplified*, since there are fewer lending opportunities to financial intermediaries than there are ultimate borrowers. In addition, the assessment of the opportunities for lending to intermediaries is generally a simpler procedure than the assessment of the opportunities for lending to ultimate borrowers.

1.6.2 The benefits to ultimate borrowers (deficit units)

These may be summarised as follows:

- Loans will generally be available for a *longer time period* from financial intermediaries than from the ultimate lenders.
- Financial intermediaries will generally be prepared to grant loans of *larger amounts* than will ultimate lenders.
- Using financial intermediaries will generally involve *lower transaction costs* than would be incurred if borrowers had to approach ultimate lenders directly.
- The *interest rate* will generally be *lower* when borrowing from financial intermediaries, compared with borrowing directly from ultimate lenders. As we have seen, financial intermediaries, through the minimisation of information costs and the diversification of risk, can actually reduce the cost of intermediation.
- When borrowing from financial intermediaries, there is a greater likelihood that loans will be *available when required*.

1.6.3 The benefits to society as a whole

Financial intermediation is not only beneficial to borrowers and lenders, but it is considered likely to cause:

- A *more efficient utilisation of funds within an economy*, since the evaluation of lending opportunities will be improved.
- A *higher level of borrowing and lending to be undertaken*, due to the lower risks and costs associated with lending to financial intermediaries.
- An *improvement in the availability of funds to higher-risk ventures*, due to the capability of financial intermediaries to absorb such risk. High-risk ventures are widely considered to be important for creating the basis of future prosperity for an economy.

1.7 Conclusion

This chapter has analysed the key features of financial intermediation. Banks, as other financial intermediaries, play a pivotal role in the economy, channelling funds from units in surplus to units in deficit. They reconcile the different needs of borrowers and lenders by transforming small-size, low-risk and highly liquid deposits into loans which are of larger size, higher risk and illiquid (transformation function). We discussed the main reasons banks have advantages in the intermediation process relating to matching the needs of ultimate lenders (depositors) and borrowers. In particular, we explored the concepts of transaction costs, economies of scale and economies of scope.

The relevance of information costs and the notion of information asymmetries were also introduced. These are costs due to imperfect distribution of information among parties and can be defined as situations in which one or more of the parties to a transaction does not have all or part of the relevant information needed to tell whether the terms of the contract are mutually acceptable and/or are being met. Situations of this kind give rise to adverse selection and moral hazard problems; another type of information asymmetry relates to the agency costs between the principal (bank) and agent (borrower).

The reasons for the existence of banks are then related to five theories that concern: delegated monitoring; information production; liquidity transformation; consumption smoothing; and the role of banks as a commitment mechanism. The chapter concludes with an overview of the main benefits of financial intermediation.

Key terms

Adverse selection	Financial asset	Moral hazard	Surplus unit
Agency costs	Financial claim	Principal-agent problems	Transactional banking
Deficit unit	Financial liability	Relationship banking	Transaction costs
Delegated monitoring	Indirect finance	Screening	
Direct finance	Information asymmetries	Shadow banking	
Economies of scale	Liquidity risk	Signalling	
Economies of scope			

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REVISION QUESTIONS AND PROBLEMS

- 1.1 What is the role of financial intermediaries in an economy?
- 1.2 What is special about banks?
- 1.3 How do lenders' and borrowers' requirements differ? How can financial intermediaries bridge the gap between them?
- 1.4 Explain how banks can lower transaction costs.
- 1.5 Explain the relevance of information asymmetries in the intermediation process.
- 1.6 How do adverse selection and moral hazard affect the bank lending function? How can banks minimise such problems?
- 1.7 How are banks affected by agency problems?
- 1.8 Describe the main theories put forward to explain the existence of financial intermediaries.
- 1.9 Explain the concept of delegated monitoring.
- 1.10 What are the costs and benefits of financial intermediation?